

Interaction 2x2 Simulation

Meeting prep: how to read the numbers and the surface plots — clean wins, honest oddities, and discussion prompts

1. The numbers — what to point out

1a. Clean wins (lead with these)

Overfitting control works.

When M_1 fits an interaction that isn't there (Cell B), $RMSE_{f12}$ is tiny everywhere (0.020–0.067) and the 95% band covers zero at $\geq 96\%$. The adaptive $\tau^2_{s,uv}$ with $a_s=1$, $b_s=0.005$ is doing its job — no need to manually pre-screen pairs.

Main effects are untouched by the interaction machinery.

Cell A vs Cell B main-effect RMSEs agree to three decimal places in most cells. Adding interaction blocks doesn't leak into the main-effect fits. This is the non-obvious structural win — the RW2 prior on β_{uv} doesn't "borrow" curvature from the main-effect blocks.

Recovery scales gracefully.

$RMSE_{f12}$ in Cell D stays between 0.053 and 0.100 across $p \in \{2,3,4\}$ and both phases. The Phase 1 $\rightarrow$ Phase 2 degradation (e.g. $p=4$: 0.053 $\rightarrow$ 0.099) is small and expected — fitting 5 additional null blocks costs something, but the construction absorbs it.

1b. Interesting oddities (flag honestly)

Oddity 1. $p=2$ Phase 1 Cell C is suspicious — cross-DGP WAIC comparison is not meaningful.

Compare Cell A vs Cell C at $p=2$ Phase 1:

	Cell A (NULL / M_0)	Cell C (INT / M_0)
est. σ^2	0.736	0.241
est. ρ	0.305	0.131
est. τ^2	0.270	0.290
WAIC	1171.5	644.6

WAIC literally halves going from NULL to INT under the *same* model M_0 . That's only possible if the *data* differ between DGPs — each cell uses a different random draw (different spatial GP realization). **So cross-DGP WAIC comparisons aren't meaningful; only within-DGP ones are.** The WAIC comparison to report is always **Cell C vs Cell D** (same data, different model), not Cell A vs Cell C. Flag this to Sounak so he doesn't misread the table. Consider rerunning with a fixed seed across cells, or at minimum add a caption note to the tables.

Oddity 2. WAIC disagrees with RMSE at $p=3$.

At $p=3$, Phase 1: $RMSE_{f12}$ in Cell D is 0.063 (good recovery), yet WAIC prefers M_0 by 13.6. Phase 2: $RMSE_{f12}$ is 0.100, WAIC prefers M_0 by 10.0. At $n=300$ with an interaction whose true surface has range $[-0.58, 0.42]$ (amplitude ≈ 1), the p_{WAIC} penalty for adding a 49-parameter block is not offset by the log-predictive gain. This is a finite-sample phenomenon, not a failure — and it's actually a nice talking point:

WAIC is conservative about declaring interactions, which is arguably desirable for epidemiological applications where false positives on interaction terms can be costly.

Oddity 3. Coverage is 1.00 too often — bands are calibrated but conservative.

Most Cell B and Cell D cells hit exactly 1.00. Combined with non-trivial $\text{RMSE}_{f_{12}}$ of 0.05–0.07, this strongly suggests the bands are wider than optimal — conservative rather than overconfident. The cells with realistic coverage (0.900 and 0.937) are where the signal is hardest. The Phase 1 $p=4$ mean band width of 0.348 is wide relative to the ≈ 1 signal amplitude. Two non-alarming explanations: (i) RW2 priors are known to produce conservative credible bands on top-level quantities (Lang & Brezger 2004 note this); (ii) $M=8$ knots per axis may be slightly more than needed at $n=300$, inflating posterior variance. Worth mentioning; not a problem.

Oddity 4. σ^2 scatter — don't lead with this.

True $\sigma^2=0.5$; estimates range from 0.24 (Cell C, $p=2$ P1) to 0.89 (Cell A, $p=2$ P2). At $n=300$ with 6 variance parameters competing for signal, this level of scatter is expected — σ^2 is the hardest parameter to pin down in any spatial random-effect model. If asked, say “consistent with the σ^2 identifiability issues we saw at $n=400$ in Section 5.6.”

1c. The one-sentence pitch for Sounak

The construction works: it recovers true interaction surfaces with RMSE of 0.05–0.10 at $n=300$, it shrinks null interaction blocks to near-zero with conservative coverage, and adding interaction blocks doesn't disturb main-effect recovery. WAIC is conservative about declaring an interaction at $p=3$ and $n=300$, consistent with the modest amplitude of the true surface, but RMSE-based recovery is clean across all six (p , phase) combinations.

2. The surface plots — how to read them

Reading order: (a) truth → (b) recovery under correct model → (c) null-shrinkage check → (d) residual from truth. Panel numbers refer to Figure 21 ($p=4$, Phase 2).

Panel (a) — True $f(\mathbf{x})$

The truth has a clear **four-corner saddle pattern**: two positive-valued corners (bottom-left, top-right, shown red) and two negative-valued corners (top-left, bottom-right, shown blue), with the zero level-curve forming a cross through the middle. This is mathematically what $(\sin(\pi x_1) - 2/\pi)(e^{x_2-0.5} - C_2)$ produces: both factors are monotonic-ish in their arguments after centering, so their product flips sign across the two middle lines $x_1 \approx 0.5$ and $x_2 \approx 0.5$. Range $[-0.58, 0.42]$ matches theory.

Panel (b) — Cell D posterior mean (the recovery)

The four-corner structure is recovered, with three visible distortions:

- **Corners are smoothed off.** Truth corner peaks/troughs reach ± 0.4 – 0.58 ; the posterior mean reaches only ± 0.3 – 0.4 . Classic shrinkage from the RW2 prior combined with limited data at the corners of $[0,1]^2$ (covariate density is uniform, so corners have fewer neighbors).
- **Wiggles in the interior.** The zero-crossing cross is less sharp than in the truth. This is where the RMSE of 0.099 accumulates.
- **Top-left and top-right are slightly merged.** The posterior has trouble separating fine structure near $x_2=1$ — only a handful of observations sit at the boundary.

This is a textbook “reasonable recovery with boundary and corner shrinkage” result. Nothing alarming.

Panel (c) — Cell B null-shrinkage check (single most important plot)

Truth is identically zero, yet the posterior mean shows small bumps with amplitude ± 0.05 – 0.2 .

What to say:

“The fitted surface is not perfectly flat — small non-zero structure of amplitude up to ± 0.2 appears — but crucially the credible bands cover zero at 96.3% everywhere, and the RMSE is only 0.061. The (x_1, x_2) regions where the posterior mean deviates most are the corners, where there’s the least data — which is exactly where shrinkage has the least leverage.”

What Sounak might push back on:

“Are those corner bumps a real artifact of the RW2 prior?” **Answer: yes, partially.** The RW2 prior penalizes second differences, so it allows *linear trends* to persist. What you see in panel (c) is compatible with weak linear drift along x_1 and x_2 . This is a known property of RW2 priors — not a bug, but worth being aware of. If needed, one could switch to RW1 or an ICAR-type 2D penalty for stronger zero-pull, though at the cost of smoothness properties elsewhere.

Panel (d) — Cell D residual

The residual is dominated by **boundary errors** — the largest deviations are at the bottom-left corner (-0.4 blue region) and secondary structure along the right edge. The interior of the unit square has residuals within ± 0.1 .

What to say:

“The residual plot confirms that the interior of the surface is well-recovered; the 0.099 RMSE is driven by the boundary and corners. If the application cares specifically about interaction effects at extreme

exposure combinations, we should note this as a limitation — otherwise the recovery is clean.”

The combined story (60-second version)

The four panels tell a coherent story. The truth has a four-corner saddle [panel a]. Under correct specification [panel b] we recover the saddle faithfully in the interior, with expected shrinkage at the corners. Under null truth [panel c] the surface stays near zero with conservative coverage, with small boundary wiggles that reflect RW2's permissiveness toward linear trends. The Cell D residual [panel d] confirms the interior is recovered well; the RMSE of 0.099 is dominated by boundary behaviour at the corners of $[0,1]^2$. This is the expected fingerprint of tensor-product penalized splines — nothing surprising, everything in the right direction.

3. Discussion prompts — questions to raise or expect

Questions to raise proactively

- **Seed consistency across cells.** Should we rerun the 2×2 with a fixed data seed across {NULL, INT} so the WAIC differences between Cell A and Cell C are meaningful? Currently only Cell C vs D and Cell A vs B are comparable within-DGP.
- **Knot count at $n=300$.** Is $M=8$ the right choice, or should we reduce to $M=6$ to match the earlier main-effects simulations and narrow the credible bands?
- **Scaling up.** The $n=300$ setting is a proof-of-concept. Should we rerun $p=4$ Phase 2 at $n=1000$ to get the interaction-chapter headline numbers closer to the main-effects benchmark?
- **WAIC vs other criteria.** Given WAIC is conservative about interactions at $n=300$, $p=3$, should we also report LOO-CV (Vehtari et al. 2017) for consistency with current Bayesian practice?

Questions Sounak might ask — prepared answers

“Why is coverage exactly 1.00 everywhere?”

Because the 95% credible bands are wider than optimal at $n=300$ with $M=8$. Two contributing factors: (i) RW2 priors are conservative on top-level quantities (Lang & Brezger 2004); (ii) the 49-parameter interaction block is rich relative to the data. Band width at $p=4$ Phase 1 averages 0.348 — wide against a signal of amplitude ≈ 1 . Calibrated but conservative.

“Why does σ^2 swing so much across cells?”

σ^2 is the hardest parameter to pin down in any spatial random-effect model, especially at modest n with 6 variance parameters competing. Consistent with the identifiability pattern documented at $n=400$ in Section 5.6 of the main-effects simulations.

“What about the corner bumps in Cell B — is the prior broken?”

No, it's working as designed. RW2 penalizes second differences (curvature) but leaves linear trends in its null space. A weak residual linear drift on the interaction surface is compatible with the prior. Credible bands correctly cover zero at 96%, and RMSE is only 0.061. If a stronger zero-pull were desired, RW1 or an ICAR-type 2D penalty is the natural alternative — but we'd lose the smoothness properties that motivate RW2 in the first place.

“Why does WAIC pick M_0 at $p=3$ when RMSE_{f12} is fine?”

Because at $n=300$ the log-predictive gain from fitting the 49-parameter interaction block doesn't offset the p_{WAIC} penalty for adding it, when the true signal amplitude is ≈ 1 on a unit-square domain. This is a finite-sample phenomenon. WAIC being conservative about declaring interactions is actually a **desirable** property for applications where false positives on interaction effects carry real cost (e.g., epidemiology).

Two things to decide at the meeting

- **Do we rerun at $n=1000$ before the chapter is finalized?** — aligns with the main-effects simulation sample size, tightens coverage, tightens WAIC verdict.
- **Do we fix the data seed across cells?** — makes Cell A vs Cell C WAIC directly comparable, at the cost of losing across-cell variability as a sanity check.